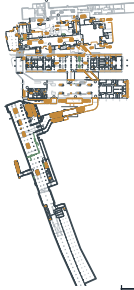


INPUT / NETWORK CONSTRUCTION

1. Station geometry

Longyang Road Station
Multi-line transfer complex



2. Evacuation demand



3. Dynamic network abstraction



ARRIVAL-TIME-AWARE ROUTE PLANNING

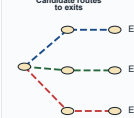
1 Dynamic state update



2 ETA-based queue forecast



3 AA* route evaluation



$$C(P) = T_{move} + W_{queue} + W_{recv} + R_{density}$$

T_{move} travel time
 W_{queue} ETA queue wait
 W_{recv} receiving delay
 $R_{density}$ density exposure
Adaptive A* search

4 Rolling route assignment



DYNAMIC EVACUATION-FLOW EXECUTION

Accepted cohorts continue; only undecided cohorts are replanned



OUTPUT / EVALUATION

1. Route allocation



2. Network route export



3. Continuous-space check



Fig. 1. Framework of the proposed arrival-time-aware evacuation planning method for Longyang Road Station.